

CVE Summary for Documented Wi-Fi Chipsets

IoT / Medical Device Cybersecurity Testbed Capstone

MediaTek MT7610UN · MediaTek MT7612UN · Realtek RTL8812AU · Realtek RTL8811AU

Research summary — August 2026

Verified CVEs for the listed chipsets (or their common drivers/firmware). Highest relevance: wireless protocol flaws (WPS / out-of-bounds writes) enabling proximal attacks or DoS.

CVE Overview Table

Chipset(s)	CVE(s)	Severity	Key Details
MT7610UN + MT7612UN (overlap)	CVE-2021-37584, -37563, -37562, -37561 (+ related WPS set incl. -37566)	High (≈ 8.2)	WPS / IEEE 1905 mishandling → OOB write/read. Proximal privilege escalation when WPS used. Affects MT7610, MT7612 + related 76xx (v7.4.0.0). MediaTek Jan 2022 bulletin.
	CVE-2022-26445, -26444, -26443, -26442, -26441	Medium (6.7)	Wi-Fi driver missing bounds check → OOB write. Local privilege escalation (System needed). Affects MT7610, MT7612 + others ($\approx$ v7.6.2.3). Aug 2022 bulletin.
RTL8812AU	CVE-2020-26652	High (7.5)	DoS in <code>n180211_send_chandef</code> (rtl8812au v5.6.4.2, often aircrack-ng driver). Network-adjacent.
RTL8811AU	CVE-2025-8301	High (≈ 7.8)	Heap buffer overflow in <code>N6CSet_DOT11_CIPHER_DEFAULT_KEY</code> (rtwlanu.sys) → local LPE to SYSTEM.
	CVE-2025-8298	Low-Med	OOB read in <code>N6CQueryInformationHandleCustomized11n0ids</code> → info disclosure (chainable).
	CVE-2025-8299, -8300, -8302	High	Related heap overflows in same Windows driver / rtl81xx SDK family → local LPE. Fixed in later builds (e.g. 1030.52.0325.2025+).

Additional Notes

- **Highest testbed value:** MediaTek WPS OOB set (2021) and 2022 driver OOB writes — strongest shared attack surface for protocol fuzzing / malformed management frames.
- Later MediaTek WLAN AP driver issues (2025 series) primarily list newer parts (MT7615+) but share driver lineage; confirm exact applicability.
- Linux `mt76` driver (common for these USB adapters) has separate kernel CVEs (use-after-free, bounds); not chipset-specific.
- Older ZDI-20-1400 (RTL8811AU, 2020): OOB write in 802.11 frame processing with potential network-adjacent RCE (no public CVE assigned in checked sources).
- Applicability depends on exact firmware/driver version and features enabled (e.g., WPS). Many require local/System access or specific conditions.

Sources: NVD, OpenCVE, MediaTek Product Security Bulletins (Jan/Aug 2022+), Realtek / ZDI advisories, aircrack-ng rtl8812au issues. Always re-verify against current NVD and vendor bulletins for the precise driver/firmware in use.